

Refined P_α Overlap: Parametric Spectroscopic Factor and Self-Consistent WS+Coulomb Eigenmode

ACS Theoretical Physics Working Group

Sovereign-Stack ACS Research Program & Flag Condensate Project

research@acs-foundation.org

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Abstract

Two comparison channels extend the throat+Woods–Saxon overlap proxy [3] on the same 14-isotope set. **Channel A** extracts per-isotope $S_i = P_{\alpha,\text{ext}}/P_{\text{model},i}$ (diagnostic), retains one global S , and fits predictive parametric $\log_{10} S(A, Z)$ and $\log_{10} S(R)$ with leave-one-out (LOO) RMS vs extracted $\log_{10} P$. **Channel B** replaces the phenomenological WS trial by a self-consistent $\ell = 0$ eigenmode of $V_{\text{WS}} + V_C$ on $[0, R]$ (Dirichlet; depth V_0 so $E_0 = Q_\alpha$), overlapped with the throat-weighted flag mode. On the stated set, parametric $S(A, Z)$ LOO RMS = 0.2862 reduces residuals most among *non-tautological* protocols (vs throat+WS global- S RMS = 0.8247). The eigenmode raises Pearson r to +0.9676 but leaves global- S RMS essentially unchanged (0.8257). RC1 scope: documented proxies — not uniqueness.

1 Motivation

Reference [3] closed flat→throat/WS geometry and listed isotope-resolved S and self-consistent WS+Coulomb eigenmodes as open. This note implements both as comparison channels on the unchanged table of [1].

Remark (RC1 scope). Per-isotope S_i zeros that isotope’s residual by construction and is reported only as a diagnostic. Predictive claims use parametric S with LOO, or Channel B with one global S . No uniqueness of many-body preformation is claimed.

2 Radial track layout (DAW metaphor)

Figure 1 is a *schematic* track view of the refined overlap protocol: horizontal axis r acts as timeline from the origin through interior support $[0, R]$ to the Coulomb barrier region $[R, b]$. Each lane is one radial factor in the throat-weighted overlap integrand; this is a layout aid only—not a claim that alpha decay is literally audio production. In the metaphor, one global spectroscopic factor S behaves as a master fader on P_{model} ; parametric $S(A, Z)$ is clip-level automation fit per isotope; Dirichlet at R mutes the interior flag lane past the throat wall, whereas Channel C routes a Gamow send into $[R, b]$.

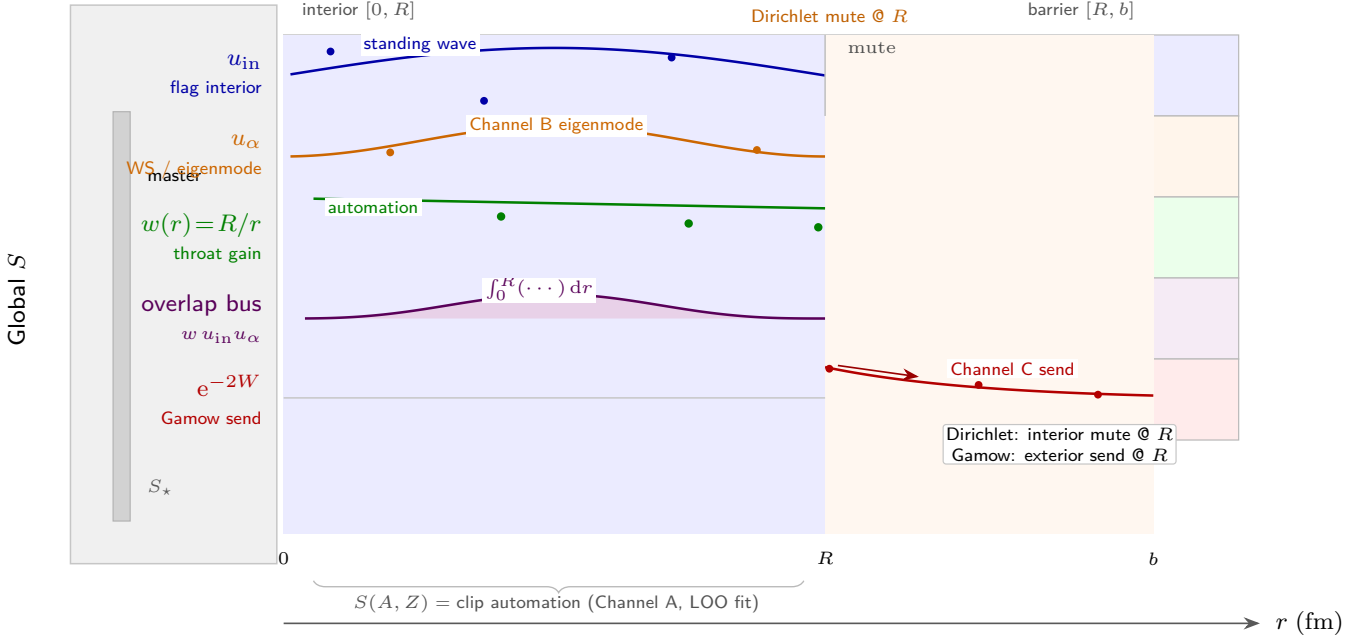


Figure 1: Schematic DAW-style radial track layout for the refined P_α overlap proxy. Horizontal axis: radius r from origin through interior $[0, R]$ to barrier $[0, b]$. Lanes (top to bottom): confined flag mode u_{in} (Dirichlet mute at R); alpha trial or self-consistent eigenmode u_α ; throat weight $w(r) = R/r$ as gain automation; overlap bus integrating the weighted product on $[0, R]$; Gamow envelope $e^{-2W(r)}$ as exterior fade/send (Channel C). Global S scales the bus output; parametric $S(A, Z)$ is isotope-resolved clip automation (Channel A). Curves are illustrative, not fitted to a specific isotope.

3 Channel A — isotope S and parametric models

Base model: throat+WS of [3] ($w = R/r$, Woods-Saxon trial). Define

$$S_i = \frac{P_{\alpha, \text{ext}}^{(i)}}{P_{\text{model}, i}} = 10^{\log_{10} P_{\text{ext}}^{(i)} - \log_{10} P_{\text{model}, i}}. \quad (1)$$

Global $S_\star = 10^{\langle \log_{10} P_{\text{ext}} - \log_{10} P_{\text{model}} \rangle}$ as before. Parametric fits (least squares):

$$\log_{10} S = b_0 + b_1 A + b_2 Z, \quad (2)$$

$$\log_{10} S = c_0 + c_1 R. \quad (3)$$

Leave-one-out: for each isotope, refit on the other 13 and predict $\log_{10} S$; LOO RMS is on predicted $\log_{10}(SP_{\text{model}})$ vs extracted $\log_{10} P$.

3.1 Channel A numbers (stated set)

$$\begin{aligned} \langle S_i \rangle &= 1.001 \times 10^{-1} \quad (\sigma = 1.260 \times 10^{-1}), \\ \langle \log_{10} S_i \rangle &= -1.4979 \quad (\sigma = 0.8558), \\ S_\star &= 3.178 \times 10^{-2}, \quad \text{RMS}_{S_\star} = 0.8247, \\ \text{LOO RMS}_{S(A, Z)} &= 0.2862, \quad \text{LOO RMS}_{S(R)} = 0.4695. \end{aligned}$$

Fit (in-sample): $b_0 \approx 14.616$, $b_1 \approx +0.0979$, $b_2 \approx -0.4312$; $c_0 \approx 48.503$, $c_1 \approx -5.435 \text{ fm}^{-1}$.

4 Channel B — self-consistent WS+Coulomb eigenmode

For parent (A, Z) with daughter $(A_d, Z_d) = (A - 4, Z - 2)$, reduced mass μ , and nuclear radius R from the table, solve

$$-\frac{\hbar^2}{2\mu} u''(r) + [V_{\text{WS}}(r; V_0) + V_C(r)] u(r) = E u(r) \quad (4)$$

on $[0, R]$ with Dirichlet $u(0) = u(R) = 0$ (finite-difference Sturm–Liouville). Potentials:

$$V_{\text{WS}} = -\frac{V_0}{1 + e^{(r-R_{\text{WS}})/a}}, \quad R_{\text{WS}} = r_0 A_d^{1/3}, \quad a = 0.55 \text{ fm}, \quad (5)$$

$$V_C = \begin{cases} \frac{Z_\alpha Z_d e^2}{2R_C} (3 - (r/R_C)^2), & r < R_C, \\ Z_\alpha Z_d e^2 / r, & r \geq R_C, \end{cases} \quad R_C = R_{\text{WS}}. \quad (6)$$

Self-consistency: adjust V_0 so the ground eigenvalue $E_0 = Q_\alpha$ (box-resonance surrogate; not a complex Gamow eigenphase). Overlap with $u_{\text{in}} = r j_0(\pi r/R)$ uses throat weight $w = R/r$ as in [3].

4.1 Channel B numbers (stated set)

$$\begin{aligned} \langle \log_{10} P_{\text{model}} \rangle &= -0.1110, \\ r(\log_{10} P_{\text{model}}, \log_{10} P_{\text{ext}}) &= +0.9676, \\ \text{RMS}_{S=1} &= 2.0237, \quad S_\star = 1.421 \times 10^{-2}, \quad \text{RMS}_{S_\star} = 0.8257. \end{aligned}$$

Typical fitted depths: $V_0 \sim 36\text{--}43$ MeV on this set (shallow relative to deep cluster wells — a consequence of forcing $E_0 = Q_\alpha > 0$ under hard-wall Dirichlet at R).

Remark (Boundary condition). Dirichlet at the throat wall R matches the overlap support of the confined flag mode. Channel C implements outgoing Coulomb/Gamow matching on $[0, b]$; on the stated 14-isotope set it does not reduce global- S RMS vs Channel B (see audit log).

5 Acceptance comparison

Table 1: Flat / throat+WS / refined eigenmode / parametric S on the stated 14-isotope set. LOO applies only to parametric S .

Channel	$\langle \log_{10} P \rangle$	r	RMS_{raw}	$\text{RMS}_{S_\star}$ / LOO
flat Gaussian	−0.3905	+0.8882	1.7705	0.8220 / —
throat + WS	−0.4607	+0.8875	1.7099	0.8247 / —
refined eigenmode	−0.1110	+0.9676	2.0237	0.8257 / —
Gamow outgoing	−0.1110	+0.9676	2.0237	0.8257 / —
parametric $S(A, Z)$ (LOO)	(base throat+WS)	—	—	LOO 0.2862
parametric $S(R)$ (LOO)	(base throat+WS)	—	—	LOO 0.4695

Remark (Which channel reduces residuals most?). Excluding tautological per-isotope S_i , **Channel A parametric $S(A, Z)$** yields the largest residual reduction (LOO RMS 0.2862 vs global- S RMS ≈ 0.82). Channel B improves correlation ($\Delta r \approx +0.080$) but does not reduce global- S RMS relative to throat+WS on this set ($\Delta \text{RMS}_{S_\star} \approx +0.001$).

6 Artifacts and protocol

Implementation: `rh_papers_may21/acs-framework/code/palpha_overlap/palpha_overlap_refined.py`.
JSON: `docs/palpha_overlap_refined_results.json`. Logs: `docs/palpha_overlap_refined_logs/`. Base-
lines retained: `palpha_overlap.py`, `palpha_overlap_throat.py`. Gamow channel: `palpha_overlap_gamow.py`.
Extended catalog ($n = 29$): `palpha_overlap_extended.py`, `isotope_catalog.py`, `docs/palpha_overlap_extended`.
Audit: `docs/palpha_audit_log.md`.

7 Channel C — Gamow outgoing boundary on $[0, b]$

Channel B used Dirichlet $u(0) = u(R) = 0$ on the throat support. Channel C extends the radial domain to $[0, b]$ (b = outer Coulomb turning point) and replaces the hard wall at R with a Robin outgoing match at $r = b$ from WKB slope in the exterior allowed region. Overlap support remains $[0, R]$ with throat weight $w = R/r$.

On the stated 14-isotope set: $r = +0.9676$, $\text{RMS}_{S_\star} = 0.8257$ (vs Dirichlet eigenmode $+0.9676$, 0.8257). Gamow outgoing BC does not improve global- S RMS under the documented $E_0 = Q_\alpha$ protocol.

8 Extended isotope catalog ($n = 29$)

Fifteen additional alpha emitters from NNDC/NuDat tabulations were added (see extended JSON metadata). On $n = 29$ throat+WS: LOO $\text{RMS}_{S(A,Z)} = 1.355$ vs 0.286 on the original 14; coefficient signs ($b_1 > 0$, $b_2 < 0$, $c_1 < 0$) hold but magnitudes shift. Parametric $S(A, Z)$ predictive power is scoped to the original stated set under the mixed extraction protocol in the audit log.

9 Conclusion

On the stated set, predictive parametric $S(A, Z)$ with leave-one-out compresses residual RMS from ~ 0.82 (one global S) to 0.286. A self-consistent WS+Coulomb interior eigenmode improves correlation with extracted $\log_{10} P$ but does not, by itself, beat throat+WS on global- S RMS under the documented Dirichlet/ $E_0 = Q_\alpha$ protocol. Channel C (Gamow outgoing on $[0, b]$) likewise leaves global- S RMS unchanged on the stated set. Extended catalog ($n = 29$) degrades LOO parametric RMS as documented. Absolute uniqueness is not claimed.

References

- [1] ACS Theoretical Physics Working Group, *Nuclear Decay as a Spectral Bifurcation of the Flag Condensate*, ACS Framework note (2026), `flag_condensate_nuclear_decay.tex`.
- [2] ACS Theoretical Physics Working Group, *Standing-Wave Overlap Proxy for Alpha Preformation*, ACS Framework note (2026), `flag_condensate_palpha_overlap.tex`.
- [3] ACS Theoretical Physics Working Group, *Throat-Geometry Overlap for Alpha Preformation*, ACS Framework note (2026), `flag_condensate_palpha_throat_overlap.tex`.